technische universität dortmund

Multilevel model for longitudinal data

Analysing Growth Trajectories of Asian Children Using Repeated Weight Measurements

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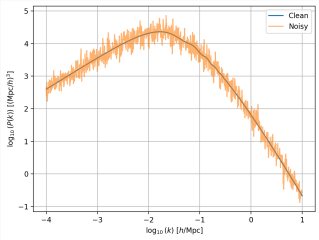
Introduction

Dataset Overview

Asian Children's Weight Measurements

1. Contains repeated weight measurements of 568 Asian children living in a British community, recorded at up to five clinic visits over time
2. Ages correspond approximately to target examination dates at 6 weeks, 8 months, 12 months, and 27 months. (In days)



https://wallpapercave.com/w/wp8201987

Dataset Overview

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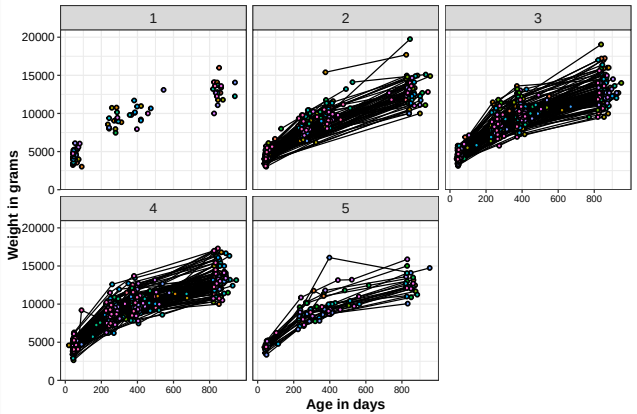
Variable	Description
ChildID	Unique identifier for each child
Age	Age in days when measured
Weight	Recorded weight in grams
Gender	1 = Boy; 2 = Girl

Dataset Overview

Variables

- ChildID
- Age (in days, metric)
- Weight (in grams)
- Gender
  (binary: Boy = 1, Girl = 2)



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Motivation

Why Use Multilevel?

Goal : To understand how children's weight changes over time and how those growth patterns differ between individuals.

1. Repeated measurements $\rightarrow$ observations are not independent.
2. Individual differences in growth $\rightarrow$ Children start at different baseline weights and grow at different rates.
3. Unequal number of observations & irregular timing $\rightarrow$ Some children have fewer visits or varying ages at measurement

"Measurements are nested within children — requiring a model that captures both levels of variation"

Two-Level Structure

Level 1 – Within-Person Level

- Represents repeated measurements over time for each child.
- Outcome measured at each occasion:
  - Weight(log_Weight) against Age (log_Age)
- Captures *within-child variation* (how a single child's weight changes with age).

Level 2 – Between-Person Level

- Represents individual children as grouping units.
- Variables constant for each child:
  - Gender
- Captures *between-child variation* (differences in overall weight levels and growth rates).

Building Multilevel Models

Classical Regression Model

Model :

$$\log(\text{Weight}_{it}) = \beta_0 + \beta_1 \log(\text{Age}_{it}) + \beta_{gen}\text{Gender}_{it} + e_{it}$$

- β_0: expected log(weight) when $\log(\text{Age}) = 0$ and Gender = 0
- β_1: expected change in log(weight) for a one-unit increase in $\log(\text{Age})$
- β_{gen}: average difference in log(weight) between genders, holding age constant
- e_{it}: error term for child i at time t

Assumption :

$$e_{it} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Random Intercept Model

Model :

$$\log(\text{Weight}_{it}) = \gamma_{00} + U_{0i} + e_{it}$$

- γ_{00}: grand mean intercept across all children
- U_{0i}: random intercept — child-specific deviation from baseline
- e_{it}: residual within-child variation at time t

Assumptions :

$$U_{0i} \sim \mathcal{N}(0, \tau_{U0}^2), \quad e_{it} \sim \mathcal{N}(0, \sigma^2), \quad U_{0i} \perp e_{it}$$

Purpose → allows each child to have their own baseline.

Random Intercept and Random Slope Model

Model :

$$\log(\text{Weight}_{it}) = (\gamma_{00} + U_{0i}) + (\gamma_{10} + U_{1i}) \log(\text{Age}_{it}) + \gamma_{01}\,\text{Gender}_i + e_{it}$$

- γ_{00}: grand mean intercept across all children
- γ_{01}: fixed effect of gender on baseline log(weight)
- γ_{10}: fixed slope — average rate of change in log(weight) per unit increase in log(age)
- U_{0i}: random intercept — child-specific deviation from baseline
- U_{1i}: random slope — child-specific deviation in growth rate
- e_{it}: residual within-child variation at time t

Assumptions :

$$U_{0i} \sim \mathcal{N}(0, \tau_{U0}^2), \quad U_{1i} \sim \mathcal{N}(0, \tau_{U1}^2), \quad e_{it} \sim \mathcal{N}(0, \sigma_e^2), \quad U_i \perp e_{it}.$$

Purpose → allows children to differ in baseline weight and growth rate, while also accounting for gender-specific differences in baseline size.

Results

Data preprocessing

Three different strategies to ensure numerical stability and to avoid influences of different scales on the model estimation:

- Centralization (subtracting general mean over time and individuals);
- Logarithm of the values;
- Normalization (subtracting general mean over time and individuals and dividing by standard deviation).

Random Intercept Fixed Slope model

The only model that can be estimated on the data without being too simple is the following:

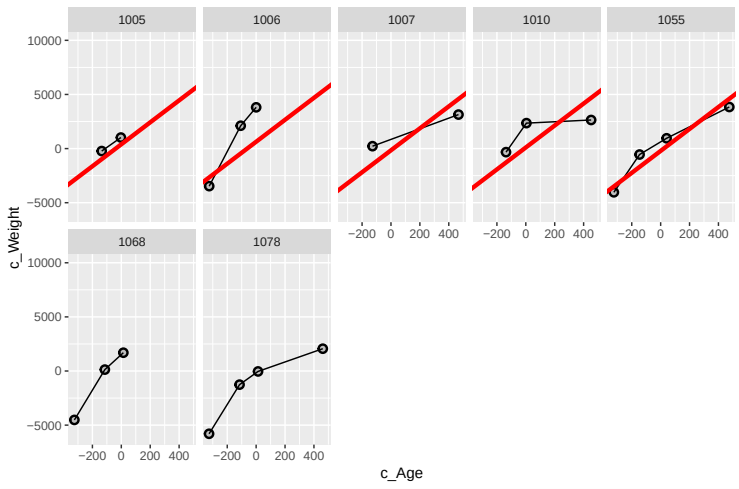
$$\overline{Weight}_{it} = \beta_0 + U_{0i} + \beta_1 \overline{Age}_{it} + \beta_2 Birthweight_i,$$

where $\overline{X}$ stands for centralized variables. All other variation either deliver the error of feature matrix being singular and therefore non-invertible.

Outlook here: bring other model configurations to work, maybe with different preprocessing steps. Also we could try to include the *Birthweight* variable as *Weight* variable values from age 0.

Result visualization

Here you can see 7 models for 7 random children that satisfy the model equation above. Here we can definitely see that the built model is underfitting for the data.



Variance Matrices: Conceptual View

Two sources of variability

- G matrix (between-person variability)
  - Variances and covariances of random intercepts and slopes
  - Captures how people differ in starting points and change rates
- R matrix (within-person variability)
  - Residual variances and covariances across time points
  - Captures fluctuations within the same individual over time

Total variability:

$$V = G + R$$

Key idea:

Observed variability comes from both between-person differences and within-person fluctuations.

Restricted Maximum Likelihood (REML)

Key idea:

- REML estimates variance components (random effects) more accurately
- Adjusts for the loss of degrees of freedom from estimating fixed effects first
- Standard choice for multilevel models
- Alternative: Maximum Likelihood (ML), but may underestimate variance when random effects are included

Intuitive visualization:

Iterative estimation process:

Fixed effects $\rightarrow$ Residuals $\rightarrow$ Random effects $\rightarrow$ Update fixed effects ...

Understanding Model Output

Example output:

Parameter	Meaning
$\gamma_{00} = 10.27$	Predicted score at Time 0
$\gamma_{10} = 1.72$	Average yearly increase
$\tau^2_{U0} = 4.10,\ \tau^2_{U1} = 0.91,\ \sigma^2_e = 0.70$	Variance components

Visual interpretation:

- **Fixed effects** $(\gamma_{00}, \gamma_{10})$ define the *group mean trajectory*
- **Random effects** (U_{0i}, U_{1i}) create individual deviations around this mean
- Graphically: the mean line with individual trajectories diverging above/below it

Key takeaway:

Fixed effects describe the average trajectory; random effects describe individual variations around it.

Advantages of Multilevel Models for Real Data

Key strengths:

- ? Handles missing or unequal timing of observations naturally
- ? Works with unbalanced datasets (different numbers of observations per person)
- ? Can include predictors at multiple levels
  - Example: gender affecting slope, age affecting intercept
- ? Captures complex covariance patterns automatically

Software options:

- SPSS: Mixed Models
- R: `lme4` package
- Python: `statsmodels` MixedLM

Beyond Linear Change

Other types of growth models:

- Polynomial growth models (quadratic, cubic) for accelerating or decelerating trends
- Nonlinear trajectories capturing more complex patterns over time
- Piecewise/discontinuous models for interventions or events

Visual intuition:

- Curved trajectories vs straight lines of linear models
- Allows modeling of real-world patterns where change is not constant

Optional conceptual extension:

- Cross-level interactions: person-level traits predicting the rate of change
- Example: age or baseline ability moderating growth trajectory

Key Points Recap

Main ideas:

- ? Longitudinal data involve repeated measures nested within individuals → need multilevel approach
- ? Multilevel models separate variation into within-person and between-person parts
- ? Fixed effects capture group trends; random effects capture individual differences in those trends
- ? These models handle missing data, irregular timing, and complex patterns flexibly

Concluding statement:

Multilevel modeling provides a powerful framework to understand both general patterns and personal trajectories in longitudinal research.